

STATS 7022 Data Science PG Formula Sheet

Secary. M

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1 Linear Regression

Models

Bias of $\hat{f}$ $b_f = \mathbb{E}[\hat{y}_i] - y_i$

Bias-Var Trade-off $\mathbb{E}[(y_0 - \hat{y}_0)^2] = \text{Var}(\hat{y}_0) + \text{Bias}(\hat{y}_0)^2 + \text{Var}(\epsilon)$

TSS $= \sum_{i=1}^n (y_i - \bar{y})^2$ **RSS** $= \sum_{i=1}^n (y_i - \hat{y}_i)^2$

MSE (Loss) $= \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$ **F** $= \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n-p-1)}$

R² $= 1 - \frac{\text{RSS}}{\text{TSS}}$ **RMSE** $= \sqrt{\frac{\text{RSS}}{n-p-1}}$

2 Classification

ROC

	Condition True	Condition False
Prediction True	True Positive (TP)	False Positive (FP)
Prediction False	False Negative (FN)	True Negative (TN)

Box-Cox Transformation

$$y = \begin{cases} \frac{x^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \log(x), & \lambda = 0 \end{cases}$$

Naive Bayes

$$P(Y = k | X = x) = \frac{1}{Z} P(Y = k) \prod_{i=1}^p P(X_i = x_i | Y = k)$$

3 Model Selection

Subset Selection

Given model M_0 for null model, M_p for full model, and M_k for k predictors,

- **Best Subset Selection:** For $k = 1, 2, \dots, p$, fit all $\binom{p}{k}$ models and choose the best one.
- **Forward Stepwise Selection:** Start with M_0 and add one variable at a time.
- **Backwards Stepwise Selection:** Start with M_p and remove one variable at a time.

Ridge and Lasso

Ridge Regression

$$\min \left\{ \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \hat{\beta}_j^2 \right\}$$

Lasso Regression

$$\min \left\{ \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda |\hat{\beta}_j| \right\}$$

$\text{RSS}(\lambda) = (y - X\hat{\beta})^T (y - X\hat{\beta}) + \lambda \beta^T \beta$, solving gives $\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$

4 Polynomial Regression and Spline

Polynomial Regression

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \dots + \beta_d x_i^d + \epsilon_i$$

Giving K knots, we have $K+1$ intervals.

$$C_0(X) + C_1(X) + \dots + C_{K-1}(X) + C_K(X) = 1.$$

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \dots + \beta_K C_K(x_i) + \epsilon_i$$

- Freedom of Cubic Spline: $K+4$ degrees of freedom.
- A degree-d spline is a piecewise degree-d polynomial with continuity in derivatives up to degree $d-1$.
- To fit a cubic spline with K knots, we fit a linear regression with an intercept and the $K+3$ predictors: $x, x^2, x^3, h_1(x, \xi_1), h_2(x, \xi_2), \dots, h_K(x, \xi_K)$.

MLR

Linear Model: $Y = X\beta + \epsilon$, with $\mathbb{E}[\epsilon] = 0$ and $\text{Var}(\epsilon) = \sigma^2 I_{n \times n}$

Estimator of β : $Q(\beta) = \|y - X\beta\|^2$

Estimator of Variance σ^2 : $s_e^2 = \frac{1}{n-p-1} \|y - X\hat{\beta}\|^2$

Slope: $\hat{\beta} = (X^T X)^{-1} X^T y$

Simple LR: $\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

Sensitivity (Recall) $= \frac{TP}{TP + FN}$

FPR $= \frac{FP}{FP + TN}$

Specificity $= \frac{TN}{TN + FP}$

TPR $= \frac{TP}{TP + FN}$

Precision $= \frac{TP}{TP + FP}$

F1-Score $= \frac{2TP}{2TP + FP + FN}$

LDA & QDA

LDA: $\delta_k(x) = \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$

LDA ($p > 1$): $\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log(\pi_k)$

QDA: $\delta_k(x) = -\frac{1}{2} x^T \Sigma_k^{-1} x + x^T \Sigma_k^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma_k^{-1} \mu_k - \frac{1}{2} \log |\Sigma_k| + \log(\pi_k)$

Metrics

$$C_p = \frac{1}{n} (\text{RSS} + 2p\hat{\sigma}^2)$$

$$C'_p = \frac{\text{RSS}}{\hat{\sigma}^2} + 2p - n$$

$$AIC = 2p - 2\log(\hat{L})$$

$$AIC_c = AIC + \frac{2p(p+1)}{n-p-1}$$

$$BIC = \log(n)p - 2\log(\hat{L}).$$

$$\text{Adjusted } R^2 = 1 - \frac{\text{RSS}/(n-p-1)}{\text{TSS}/(n-1)}$$

LOOCV

$$\text{MSE}_i = (y_i - \hat{y}_i)^2 \quad \text{CV}_{(n)} = \frac{1}{n} \sum_{i=1}^n \text{MSE}_i,$$

$$\text{CV}_{(n)} = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{1 - h_i} \right)^2, \quad h_i = \text{diag}(X(X^T X)^{-1} X^T)$$

Cubic Spline

A cubic spline with K knots,

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \dots + \beta_K C_K(x_i) + \epsilon_i.$$

Truncated polynomial basis functions:

$$h(x, \xi) = (x - \xi)_+^3 = \begin{cases} (x - \xi)^3, & x > \xi \\ 0, & x \leq \xi \end{cases}$$

- A natural cubic spline with K knots uses K degrees of freedom

Smoothing Splines

We can try and ensure that $g(x)$ is smooth by choosing $g(x)$ that minimizes $\sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int g''(x)^2 dx$.

Effective Degrees of Freedom: $df_\lambda = \sum_{i=1}^n [S_\lambda]_{ii}$, the trace of S_λ .

5 KNN

KNN Regression

$$\hat{P}(Y=j|X=x_0)=\frac{1}{K}\sum_{i\in\mathcal{N}_0}I(y_i=j)$$

$$\hat{f}(x_0)=\frac{1}{K}\sum_{i\in\mathcal{N}_0}y_i$$

Nadaraya–Watson Kernel-

Weighted Average

$$\hat{f}(x_0)=\frac{\sum_{i=1}^nk_\lambda(x_0,x_i)y_i}{\sum_{i=1}^nk_\lambda(x_0,x_i)}$$

Epanechnikov quadratic kernel

$$K_\lambda(x_0,x)=D\left(\frac{|x-x_0|}{\lambda}\right)$$

$$D(t)=\begin{cases}\frac{3}{4}(1-t^2), & |t|<1 \\ 0, & \text{otherwise}\end{cases}$$

6 MARS and CART

MARS

$$\hat{f}(x)=\sum_{i=1}^kc_iB_i(x)$$

where $B_i(x)$ takes one of three forms:

- an intercept,
- a hinge function $\max(0,x-c)$ or $\max(0,c-x)$
- a product of hinge functions $h_j(x)h_k(x)$ for some ξ_j and ξ_k .

GCV

$$\text{GCV}=\frac{\frac{1}{N}\sum_{i=1}^n(y_i-\hat{y}_i)^2}{\left(1-\frac{ENP}{N}\right)^2}$$

where $ENP=k+\frac{d(k-1)}{2}$ and penalty term d is usually 2 or 3.

Michaelis-Menten Model

$$f(x,(K,V_m))=\frac{V_mx}{K+x}$$

Classification Trees

$$\hat{p}_{mk}=\frac{1}{N_m}\sum_{i\in R_m}I(y_i=k),\quad k(m)=\operatornamewithlimits{argmax}_k\hat{p}_{mk}$$

Misclassification rate

$$\frac{1}{N_m}\sum_{x_i\in R_m}I(y_i\neq k(m))=1-\hat{p}_{mk(m)}$$

Gini Index

$$\sum_{k=1}^K\hat{p}_{mk}(1-\hat{p}_{mk})=1-\sum_{k=1}^K\hat{p}_{mk}^2$$

Cross entropy / deviance

$$-\sum_{k=1}^K\hat{p}_{mk}\log(\hat{p}_{mk})$$

7 Random Forest

Bootstrap

Consider a random sample $S=\{X_1,X_2,...,X_n\}$, and its bootstrap resample:

$$S_b^*=\{X_{b1}^*,X_{b2}^*,...,X_{bn}^*\},$$

where

$$\bar{X}^*=\hat{\mathbb{E}}[X^*]=\frac{1}{B}\sum_{b=1}^BX_b^*$$

$$\hat{\text{var}}(X^*)=\frac{1}{B-1}\sum_{b=1}^B(X_b^*-\bar{X}^*)^2$$

$$\bar{b}_T(\theta)(\mathbf{Bias})=\hat{X}^*-X$$

Bagging

Consider $\{X_1,X_2,...,X_n\}$, $X_i\sim N(\mu,\sigma^2)$. Then,

$$\text{var}(\bar{X})=\frac{\sigma^2}{n}$$

$$\text{var}(\bar{X})=\rho\sigma^2+\frac{1-\rho}{n}\sigma^2\quad\text{with correlation }\rho$$

Guidelines

- **Classification:** Default for m is $\lfloor\sqrt{p}\rfloor$ and minimum node size of 1
- **Regression:** Default for m is $\lfloor\frac{p}{3}\rfloor$ and minimum node size of 1

SVM

P-dimensional Hyperplane

$$\beta_0+\beta_1x_1+\beta_2x_2+...\beta_px_p=0$$

Properties

$$yi(\beta_0+\beta_1x_1+\beta_2x_2+...\beta_px_p)>0$$

Inner Product

$$\langle x_i,x_{i'}\rangle=x_i^Tx_{i'}=\sum_{j=1}^px_{ij}x_{i'j}$$

Kernel functions

Linear

$$k(x_i,x_{i'})=\sum_{j=1}^px_{ij}x_{i'j}$$

Polynomial

$$k(x_i,x_{i'})=\left(1+\sum_{j=1}^px_{ij}x_{i'j}\right)^d$$

Radial

$$k(x_i,x_{i'})=\exp\left(-\gamma\sum_{j=1}^p(x_{ij}-x_{i'j})^2\right)$$

8 PCA

Consider p predictors $X_1, X_2, \dots, X_p$,

First PC

$$Z_1 = \Phi_{11}X_1 + \Phi_{21}X_2 + \dots + \Phi_{p1}X_p, \quad \sum_{j=1}^p \Phi_{j1}^2 = 1$$

$$Z_1 = \alpha_1^T X$$

$$\text{var}(\alpha_1^T X) = \alpha_1^T \sum \alpha_1$$

Lagarange

$$\alpha_1^T \sum \alpha_1 - \lambda(\alpha_1^T \sum \alpha_1 - 1)$$

Differentiating w.r.t. α_1 and setting to zero, $\Rightarrow \sum \alpha_1 = \lambda \alpha_1$

$$\Rightarrow \text{Max} \alpha_1^T \sum \alpha_1 = \text{Max} \alpha_1^T \lambda \alpha_1 = \text{Max} \lambda \alpha_1^T \alpha_1 = \text{Max} \lambda$$

9 K-means Clustering

Notation

Let $C_1, \dots, C_k$ denote sets containing indices of each observation in each cluster.

- $C_1 \cup C_2 \cup \dots \cup C_k = \{1, 2, \dots, n\}$
- $C_k \cap C_{k'} = \emptyset$ for all $k \neq k'$.

Clustering

$$\min_{C_1, C_2, \dots, C_k} \left\{ \sum_{k=1}^K \frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2 \right\}$$

Note that

$$\frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2 = 2 \sum_{i \in C_k} \sum_{j=1}^p (x_{ij} - \bar{x}_{kj})^2,$$

where

$$\bar{x}_{kj} = \frac{1}{|C_k|} \sum_{i \in C_k} x_{ij}.$$

10 Multi-dimensional Scaling

Scaling Procedures

- Classical scaling: dissimilarities are Euclidean distances.
- Ordinal scaling: uses only the order of the dissimilarity measure.

Dissimilarity Coefficients

Euclidean

$$d_{rs} = \left(\sum_{i=1}^p (x_{ri} - x_{si})^2 \right)^{\frac{1}{2}}$$

Manhattan

$$d_{rs} = \sum_{i=1}^p |x_{ri} - x_{si}|$$

Minkowski

$$d_{rs} = \left(\sum_{i=1}^p |x_{ri} - x_{si}|^R \right)^{\frac{1}{R}}$$

Association Table

	Subject r	
Subject s	1	0
1	a	b
0	c	d

Simple matching coefficient

$$\frac{a+d}{a+b+c+d}$$

Hamming distance

$$b+c$$

Jaccard's coefficient

$$\frac{a}{a+b+c}$$

Classical Scaling

Consider

$$B = XX^T$$

The (r, s) th term of B

$$b_{rs} = \sum_{j=1}^p x_{rj} x_{sj}.$$

Euclidean Distances

$$d_{rs}^2 = \sum_{j=1}^p (x_{rj} - x_{sj})^2 = b_{rr} + b_{ss} - 2b_{rs}.$$

Summing Over r, s , and Both

$$\sum_{r=1}^n d_{rs}^2 = T + nb_{ss}$$

$$\sum_{s=1}^n d_{rs}^2 = nb_{rr} + T$$

$$\sum_{r=1}^n \sum_{s=1}^n d_{rs}^2 = 2nT$$

where

$$T = \sum_{r=1}^n b_{rr},$$

i.e., the trace of B .

Rearranging gives

$$b_{rs} = -\frac{1}{2} [d_{rs}^2 - d_{r\bullet}^2 - d_{s\bullet}^2 + d_{\bullet\bullet}^2].$$

Second PC

$$\text{Max} \alpha_2^T \sum \alpha_2, \quad \text{uncorrelated with } \alpha_1^T X.$$

$$\text{cov}(\alpha_1^T X, \alpha_2^T X) = \alpha_1^T \sum \alpha_2 = \alpha_2^T \sum \alpha_1 = \alpha_2^T \lambda \alpha_1 = \lambda \alpha_2^T \alpha_1$$

Lagarange

$$\alpha_2^T \sum \alpha_2 - \lambda(\alpha_2^T \sum \alpha_2 - 1) - \phi(\alpha_2^T \alpha_1 - 0)$$

Differentiating w.r.t. α_2 and setting to zero,

$$\Rightarrow \sum \alpha_2 - \lambda \alpha_2 - \phi \alpha_1 = 0.$$

$$\Rightarrow \alpha_1^T \sum \alpha_2 - \lambda \alpha_1^T \alpha_2 - \phi \alpha_1^T \alpha_1 = 0,$$

$$\Rightarrow \phi = 0$$

$$\Rightarrow \sum \alpha_2 = \lambda \alpha_2.$$

Algorithm

1. Randomly assign a number of 1 to K to each observation.
2. Iterate until no changes in cluster assignment:
 - (a) For each cluster, calculate the centroid. The centroid is the vector of the sample mean of the p features for observations in the cluster.
 - (b) Assign each observation to the cluster of the centroid that is closest.

Dissimilarity Coefficients

The dissimilarity coefficient between two points s and r must satisfy the following constraints:

1. $d_{sr} \geq 0$ for every r and s .
2. $d_{rr} = 0$ for every r .
3. $d_{sr} = d_{rs}$ for every r and s .
4. $d_{rt} + d_{ts} \geq d_{rs}$ for every r, s and t (Only if A metric satisfies 1-3).

11 The EM Algorithm

Example Using Two Normals

Consider a model with two normally distributed groups. For ease of explanation assume that each group has the same variance.

$$\begin{aligned} Y_1 &\sim N(\mu_1, \sigma^2), \\ Y_1 &\sim N(\mu_1, \sigma^2), \text{ and} \\ Y &= (1 - \Delta)Y_1 + \Delta Y_2, \end{aligned}$$

where $\Delta \in \{0, 1\}$ and

$$P(\Delta = 1) = \pi.$$

Mixture Density

$$f(y; \pi, \mu_1, \mu_2, \sigma^2) = (1 - \pi)f_1(y; \mu_1, \sigma^2) + \pi f_2(y; \mu_2, \sigma^2)$$

where f_1 and f_2 are the respective PDFs of the normal distributions.

Log-likelihood

$$\ell(\theta; y) = \sum_{i=1}^n \log((1 - \pi)f_1(y_i; \mu_1, \sigma^2) + \pi f_2(y_i; \mu_2, \sigma^2))$$

Latent Variable

Let Δ_i be latent variables such that if $\Delta_i = 1$ then observation $i \sim \mathcal{N}(\mu_2, \sigma^2)$.

With this missing information, the likelihood becomes easier to deal with:

$$f(y_i; \pi, \mu_1, \mu_2, \sigma^2, \Delta_i) = [(1 - \pi)f_1(y_i; \mu_1, \sigma^2)]^{1 - \Delta_i} [\pi f_2(y_i; \mu_2, \sigma^2)]^{\Delta_i}.$$

Latent Variable Log-likelihood

$$\begin{aligned} \ell(\theta; \mathbf{y}, \Delta) &= \sum_{i=1}^n (1 - \Delta_i) \log((1 - \pi)f_1(y_i; \mu_1, \sigma^2)) + \Delta_i \log(\pi f_2(y_i; \mu_2, \sigma^2)) \\ &= \sum_{i=1}^n (1 - \Delta_i) \log(1 - \pi) + (1 - \Delta_i) \log(f_1(y_i; \mu_1, \sigma^2)) \\ &\quad + \sum_{i=1}^n \Delta_i \log(\pi) + \Delta_i \log(f_2(y_i; \mu_2, \sigma^2)) \end{aligned}$$

Responsibilities

Since we don't know Δ_i , we use the expected value

$$\gamma_i(\theta) = \mathbb{E}[\Delta_i | \theta, \mathbf{y}] = \mathbb{P}[\Delta_i = 1 | \theta, \mathbf{y}].$$

EM Algorithm for Two-component Normal Distribution

1. Make initial guesses of parameters: $\hat{\pi}, \hat{\mu}_1, \hat{\mu}_2, \hat{\sigma}_1^2, \hat{\sigma}_2^2$.

2. *Expectation step*: compute the responsibilities:

$$\hat{\gamma}_i = \frac{\hat{\pi} f(y_i; \hat{\mu}_2, \hat{\sigma}^2)}{(1 - \hat{\pi}) f(y_i; \hat{\mu}_1, \hat{\sigma}^2) + \hat{\pi} f(y_i; \hat{\mu}_2, \hat{\sigma}^2)}$$

3. *Maximisation step*: estimate the parameters:

$$\begin{aligned} \hat{\mu}_1 &= \frac{\sum_{i=1}^n (1 - \hat{\gamma}_i) y_i}{\sum_{i=1}^n (1 - \hat{\gamma}_i)}, \quad \hat{\mu}_2 = \frac{\sum_{i=1}^n \hat{\gamma}_i y_i}{\sum_{i=1}^n \hat{\gamma}_i}, \\ \hat{\sigma}^2 &= \frac{\sum_{i=1}^n (1 - \hat{\gamma}_i) (y_i - \hat{\mu}_1)^2 + \sum_{i=1}^n \hat{\gamma}_i (y_i - \hat{\mu}_2)^2}{n - 2}, \\ \text{and} \quad \hat{\pi} &= \frac{\sum_{i=1}^n \hat{\gamma}_i}{n}. \end{aligned}$$

4. Iterate steps 2 and 3 until convergence.